

DD: PLM-CFG-08 — Bundle Catalog Configuration

Developer implementation guide: DD_PLM-CFG-08-Adjustment_Catalog-DEV_IMPL_GUIDE-v1.0.md.

1. Purpose

This rewritten DD is the developer-facing version of the module contract
DD_PLM-CFG-08-Adjustment_Catalog-v1.0.md.

Verdict: ■ Modify (Bundle entity exists in TZ but needs full lifecycle, pricing, modification rules) **Wave:** 4 | **Section A:** Configuration

The goal of this rewrite is to make the DD easier for a mid-level developer to implement without losing the detailed source contract.

2. Implementation Scope

This DD should be implemented as part of the owning SOPHIX capability, not as an isolated service unless the deployment architecture explicitly separates that capability.

Implement this DD with these rules:

- Use the owning module APIs/repositories for authoritative writes.
- Use approved internal APIs/client wrappers when reading another module's data.
- Do not query another module's database tables directly from business flow code.
- Treat worker names as logical Camunda external-task topics, not separate deployments.
- One worker application may handle multiple topics, but metrics, retries, and logs must remain visible per topic.
- Emit success events only after the authoritative state commit succeeds.
- Use outbox publishing for Kafka events.

3. Runtime Metadata

Item	Value
Source file	/Users/macbook/Downloads/Sophix_V3_BSS_DDs_MD_2026-05-27/09_catalogs/DD_PLM-CFG-08-Adjustment_Catalog-v1.0.md
Version	v1.0
DD type	module contract
BPMN/process keys	Not explicitly defined
Drools packages	Not explicitly defined

4. Runtime Contracts To Implement

4.1 APIs

- DELETE /api/bundles/{id}
- GET /api/bundles/{id}
- GET /api/bundles?status=&franchise=&page=&size=
- PATCH /api/bundles/:id
- PATCH /api/bundles/{id}
- POST /api/bundles
- POST /api/bundles/:id/activate
- POST /api/bundles/preview-price-warning
- POST /api/bundles/{id}/activate

- POST /api/bundles/{id}/retire

4.2 Tables

- No CREATE TABLE block is explicitly declared in this DD.

For each table in this DD, implement the schema with:

- a short table comment explaining what the table is for;
- column comments or migration notes explaining each field;
- constraints and indexes from the DD;
- seed data where the DD provides operator-specific sample rows.

4.3 Worker Topics

- plm-rules
- plm-session
- subscription-engine

Worker-topic guidance:

- A BPMN service task uses the topic.
- A worker application subscribes to the topic.
- The topic handler validates input variables, calls the owning module/API, writes outputs back to Camunda, and records metrics.
- Do not treat the topic string as a shell command or Kubernetes deployment name.

4.4 Events

- No domain events are explicitly declared in this DD.

Event guidance:

- Write events through the transactional outbox.
- Include operation id, aggregate id, operator code, actor, and correlation data.
- Consumers should be able to rebuild downstream state without querying workflow internals.

5. Developer Implementation Checklist

1. Add or verify database migrations and seed rows.
2. Add or verify config rows for the operator/module.
3. Implement request/command DTOs and validation.
4. Implement idempotency and in-flight operation checks where the DD requires them.
5. Implement Drools fact mapping and package deployment where rules are used.
6. Implement BPMN process, service tasks, gateways, timers, and message catches where the DD is a flow.
7. Implement worker-topic handlers using owning module APIs.
8. Implement compensation paths and manual recovery paths.
9. Emit success/rejected events through the outbox.
10. Add smoke tests for happy path, validation failure, dependency failure, and retry/idempotency.

6. Infrastructure And AWS Notes

Deploy this DD with the existing SOPHIX platform components unless the owning module requires isolation:

Concern	Recommendation
API/runtime	Run inside the owning module deployment or existing workflow API pods.

Concern	Recommendation
Workers	Consolidate low-volume topics into shared worker apps; keep per-topic metrics.
Database	Use the owning module PostgreSQL schema and migration pipeline.
Kafka	Use the foundation Kafka/MSK setup and transactional outbox.
Camunda	Deploy BPMN files through the workflow deployment pipeline.
Drools	Deploy KJAR/rule artifacts through the approved rules pipeline.
Secrets	Use the platform secret manager; on AWS prefer Secrets Manager/Parameter Store with IRSA.
Observability	Alert on stuck operations, failed workers, outbox backlog, and external dependency failures.

7. Original DD Detail Preserved

The following section preserves the detailed source contract for implementation and review.

Verdict: ■ Modify (Bundle entity exists in TZ but needs full lifecycle, pricing, modification rules)

Wave: 4 | **Section A:** Configuration

A — Target

1. What this process is about

A **Bundle** is a set of Packages sold as one marketed product — effectively a "mega-package" with its own properties that **override** the components' properties when sold as part of the Bundle. Example: a "Triple Play" bundle combines an Internet Package (normally billed monthly), a TV Package (normally monthly), and a Voice Package (normally weekly) into one offering at one price on the Bundle's own billing cycle.

The customer subscribes to the **bundle as a single entity**, not to its individual packages. Billing charges the **Bundle's price** on the **Bundle's billing cycle** using the **Bundle's wallet and tax bindings** — the components' standalone properties are not used at charge time. The components retain their standalone properties for when they are sold individually (outside this Bundle).

Modifications are all-or-nothing: a customer wanting to change one component must change the whole Bundle.

This DD covers the **configuration** of Bundles by an admin user: create, read, update, soft-delete, plus audit. There is no workflow — administrative CRUD on catalog data.

PLM scope. PLM owns the catalog of products: Services, Packages, Bundles. PLM-CFG-08 owns Bundle and BundleComposition. References to Packages are FK links within the plm database (Bundle's components are Packages in the same module).

Downstream modules (subscription-engine, fulfillment, billing, search index, etc.) consume Bundle data via Kafka events and keep their own caches in sync. PLM never calls back into them.

2. Business rules

ID	Rule	Triggered on
R-PLM-CFG-08-1	Bundle code must be unique within the catalog (across Services, Packages, and Bundles — namespace shared with PLM-CFG-01).	Create, Update
R-PLM-CFG-08-2	Bundle must have at least 2 component Packages . A bundle of one is just that Package; a bundle of zero is not a bundle.	Create, Update
R-PLM-CFG-08-3	All component Packages must be Active (not Draft, not Retired) at the time the Bundle is moved to Active state.	Activate
R-PLM-CFG-08-4	Bundle price is independent of the sum of its component Package prices — the operator sets it directly. A warning (not a block) fires if the bundle price is higher than the sum of components (operator should confirm).	Create, Update

ID	Rule	Triggered on
R-PLM-CFG-08-5	Bundle availability must be a subset of the intersection of its component Packages' availability — applied to both <code>targetFranchises</code> and <code>targetTechRegions</code> . A bundle cannot be activated in a franchise or tech region where any component is not available.	Create, Update, Activate
R-PLM-CFG-08-6	A Bundle cannot be soft-deleted while it is referenced by an active customer subscription. The admin must first migrate or retire those subscriptions.	Delete
R-PLM-CFG-08-7	A component Package cannot be soft-deleted while it is referenced by an active Bundle. The admin must first remove the Package from the Bundle (or retire the Bundle).	Delete (on Package) — coordinated with PLM-CFG-01 / SIP-01
R-PLM-CFG-08-8	Bundle <code>billingFrequencyDays</code> is a positive integer (days per cycle). It is independent of the components' frequencies — the Bundle is charged on its own cycle, not on the components'.	Create, Update

The all-or-nothing modification rule (change the whole Bundle, not individual components) is enforced at subscription-change time (FUL-04), not at catalog-config time. PLM-CFG-08 only defines what a Bundle is and what it contains.

3. Module owner and impacted modules

Role	Module	What it does in this process
Owner	<code>plm</code>	Owns Bundle + BundleComposition data, REST APIs, the bundle admin UI, the DRL rules KJAR
Impacted (downstream consumers)	<code>subscription-engine</code> (eligibility), <code>fulfillment</code> (provisioning all components at activation), <code>billing</code> (charge bundle price), search index, BSS Backoffice (catalog views)	Consume BundleCreated/Updated/Retired Kafka events; refresh their local caches
Gateways	—	None used in this process (pure CRUD)
Engines	Drools KIE	Rule evaluation on every create/update/activate/delete

No workflow, no Camunda. No external system calls. PLM-CFG-08 is pure catalog admin.

4. Foundation references

DB → FOUNDATION_DB.md · Drools → FOUNDATION_DROOLS.md · Kafka → FOUNDATION_KAFKA.md · Cache → FOUNDATION_CACHE.md · Auth → FOUNDATION_AUTH.md.

Technical design

Module placement

Layer	Where
Frontend	Shared BSS Backoffice UI (Catalog section, Bundles sub-area)
Backend	Existing <code>plm</code> module , sub-area <code>bundle-catalog/</code>
Rules	Rules Engine (Drools) — <code>plm-rules</code> KJAR (shared with PLM-CFG-01, PLM-CFG-08 adds its own DRL files)
Events	Internal Event Bus (Kafka)

Single-tenant: same codebase per country, config + rule packages loaded at startup.

`plm` module — ecosystem (PLM-CFG-08 view)

```

flowchart LR
    UI[BSS Backoffice UI<br/>Catalog section] -->|REST| PLM[plm module<br/>bundle-catalog]
    PLM -->|invoke| RE[Rules Engine]
    PLM -->|publish| EB[Kafka]
    EB --> SIP[subscription-engine cache]
    EB --> FUL[fulfillment cache]
    EB --> BIL[billing cache]
```

```
EB --> SEARCH[search index]
PLM -.FK.-> PKG[Package catalog<br/>PLM-CFG-01 / SIP-01<br/>same plm DB]
```

plm never calls customer services, billing engine, or fulfillment workflows. Bundle catalog is purely admin-facing.

Codebase

Bundle catalog adds to the existing plm module. Stack choice (Java/Spring Boot or Laravel) was made for PLM-CFG-01 and stays the same — no per-DD stack switch.

Java/Spring Boot (continuation of PLM-CFG-01)

```
modules/plm/                                ← existing module
  src/main/java/com/sophix/plm/
    service-catalog/                         ← PLM-CFG-01 (existing)
    package-catalog/                         ← SIP-01 (later)
    bundle-catalog/                         ← PLM-CFG-08 (this DD)
      {domain, repository, service, web/rest, event, exception}/
  src/main/resources/
    config/liquibase/changelog/plm/
      004_bundle.xml                         ← new for PLM-CFG-08
      005_bundle_composition.xml            ← new for PLM-CFG-08

modules/plm-rules/                          ← existing KJAR project
  src/main/resources/
    rules/
      plm-cfg-08-bundle-defaults.drl        ← R-PLM-CFG-08-? (none needed yet – left for future)
      plm-cfg-08-bundle-validation.drl     ← R-PLM-CFG-08-1, -2, -3, -4, -5, -6, -7
  src/main/java/com/sophix/plm/facts/
    BundleFact.java                         ← new fact for PLM-CFG-08
    BundleCompositionFact.java              ← new fact
    PackageAvailabilityFact.java            ← new fact
    ReferencingSubscriptionsFact.java       ← new fact
```

Laravel (if stack chosen)

```
modules/plm/                                ← existing module
  app/
    Models/
      Bundle.php                             ← new
      BundleComposition.php                  ← new
    Http/Controllers/
      BundleController.php                  ← new
    Services/
      BundleCatalogService.php              ← new
      BundleValidationService.php            ← calls KIE Server
    Events/
      BundleCreated.php / BundleUpdated.php / BundleRetired.php
  database/migrations/
    2026_05_15_000004_create_bundle_table.php
    2026_05_15_000005_create_bundle_composition_table.php
```

Foundation-specific configurations

DB (FOUNDATION_DB.md):

- Cluster: **A** (Config & Catalog) — same as PLM-CFG-01
- Database: plm — shares the database with PLM-CFG-01
- New tables (in public schema): bundle, bundle_composition
- New events on audit_log_entry (existing table)
- Messaging tables (messaging.outbox_event, messaging.inbox_event) — already created by PLM-CFG-01, shared

Kafka (FOUNDATION_KAFKA.md):

- Topics published:
 - sophix.plm.bundle.created
 - sophix.plm.bundle.updated
 - sophix.plm.bundle.retired
- Topics consumed: none
- Key format: plm:bundle:{bundleId} (e.g., plm:bundle:bnd_01HX2Y3Z4)

- eventId format: plm:bundle:{bundleId}:evt_{ulid}

Cache (FOUNDATION_CACHE.md):

- plm does not cache its own data
- Downstream caches use key prefix sophix:plm:bundle: and TTL 24h
- Sync via Kafka topics above

Drools (FOUNDATION_DROOLS.md):

- KJAR: plm-rules-{country}_{version} (shared with PLM-CFG-01)
- KIE container ID: plm-rules-ke_1.0.0 (same)
- KIE session: plm-session (same)
- New DRL files added: plm-cfg-08-bundle-validation.drl
- New facts: BundleFact, BundleCompositionFact, PackageAvailabilityFact, ReferencingSubscriptionsFact

Camunda: Not used (CRUD only, no workflow).

Frontend

Item	Detail
App	Shared BSS Backoffice UI (this DD adds Bundles to the Catalog section, alongside Services from PLM-CFG-01)
Actor	Product Manager (full CRUD), Billing Lead (full CRUD), Admin (full CRUD), any authenticated role (read-only)
Auth	Bearer JWT (see FOUNDATION_AUTH.md)
Routes	/catalog/bundles, /catalog/bundles/new, /catalog/bundles/:id, /catalog/bundles/:id/audit

Screens

Bundle list (/catalog/bundles)

- Table: code, name, status (Draft/Active/Retired), component count, price, available franchises, updatedAt, updatedBy
- Filters: status, franchise, search (substring on code/name)
- Pagination, default 20
- Actions: New Bundle button, row click → edit

Bundle create / edit (/catalog/bundles/new, /catalog/bundles/:id)

- Form fields:
- Code (text, required, unique — validated server-side via R-1)
- Name (text, required)
- Description (text)
- Status (Draft / Active / Retired — controlled by save-as-draft / activate / retire actions, not a free field)
- Composition (multi-select of Packages, min 2 — R-2; auto-suggests available Packages from PLM-CFG-01 / SIP-01)
- Price (number, required) + a warning shown if price > sum of component prices (R-4)
- Billing frequency in days (numeric input; presets: 30 / 28 / 15 / 7 / 180 / 365 — R-8). Independent of components.
- Default wallet (dropdown) — Bundle's wallet, overrides components
- Default tax group (dropdown, optional) — Bundle's tax, overrides components
- targetFranchises (multi-select; constrained by intersection of components' franchises — R-5)
- targetTechRegions (multi-select; constrained by intersection of components' tech regions — R-5)
- Live calculation panel showing: sum of component prices, bundle price, implied discount %, franchises available
- On Submit: POST /api/bundles (new) or PATCH /api/bundles/:id (edit)

- On Activate (separate button on draft bundles): POST /api/bundles/:id/activate — fires R-3

Bundle audit (/catalog/bundles/:id/audit)

- Timeline of audit_log_entry rows for this Bundle
- Each row: timestamp, actor, action (CREATE/UPDATE/ACTIVATE/RETIRE), expandable diff
- Read-only

Client-side validation

Basic input only (required fields, numeric format). Server is authoritative for all business rules (R-1 through R-7). The dev should NOT replicate Drools logic client-side.

Data model

```
classDiagram
    class Bundle {
        BundleId id
        String code
        String name
        String description
        BundleStatus status
        Money price
        Integer billingFrequencyDays
        String defaultWalletRef
        String defaultTaxGroupRef
        List-String~ targetFranchises
        List-String~ targetTechRegions
    }
    class BundleComposition {
        BundleId bundleId
        PackageId packageId
        Int sequence
    }
    class Package {
        PackageId id
        String code
        ...
    }
    Bundle "1" --> "n" BundleComposition
    BundleComposition --> Package
```

Tables (PostgreSQL, database plm, schema public)

bundle — the Bundle catalog

The Bundle's own properties override those of its component Packages at charge time. The Bundle is billed on `billing_frequency_days`, charging price (in currency), routing to `default_wallet_ref`, with tax computed via `default_tax_group_ref`. Components' equivalent properties are not used when the customer is subscribed to the Bundle.

Column	Type	Notes		
id	TEXT PK	ULID, e.g. bnd_01HX...		
code	TEXT NOT NULL	unique across plm catalog		
name	TEXT NOT NULL	display name, e.g. "Triple Play Pro"		
description	TEXT			
status	TEXT NOT NULL	DRAFT \	ACTIVE \	RETIRED
price	NUMERIC(12,2) NOT NULL	bundle price — overrides sum of components (R-4)		
currency	TEXT NOT NULL	ISO 4217 (KES, UGX, TZS, XOF, etc.)		
billing_frequency_days	INT NOT NULL	positive integer; days per billing cycle (R-8). Independent of components' frequencies. Typical: 30, 28, 15, 7, 180, 365.		
default_wallet_ref	TEXT NOT NULL	wallet binding for the Bundle — overrides components' wallets		

Column	Type	Notes		
default_tax_group_ref	TEXT NULL	tax binding for the Bundle — overrides components' tax groups; falls back to none if NULL (charges with no tax cascade)		
target_franchises	TEXT[] NOT NULL	array of franchise codes; non-empty when status = ACTIVE; must be subset of intersection of components' franchises (R-5)		
target_tech_regions	TEXT[] NOT NULL DEFAULT '{}'	optional refinement		
created_at	TIMESTAMP NOT NULL DEFAULT NOW()			
created_by	TEXT NOT NULL			
updated_at	TIMESTAMP			
updated_by	TEXT			
retired_at	TIMESTAMP NULL	soft delete		

Indexes:

- (code) WHERE retired_at IS NULL UNIQUE
- (status) for status-filtered queries
- GIN index on (target_franchises) for franchise-availability lookups

bundle_composition — Bundle → Packages many-to-many (in PLM-CFG-08 within plm DB, so FKs are OK)

Column	Type	Notes
bundle_id	TEXT NOT NULL	FK → bundle.id ON DELETE CASCADE
package_id	TEXT NOT NULL	FK → package.id (created by SIP-01)
sequence	INT NOT NULL DEFAULT 0	display order

Primary key: (bundle_id, package_id). Indexes: (package_id) for reverse lookups (R-7: is this Package referenced by any active Bundle?).

Rules (Drools)

Rule IDs match the business-rule table in Section 2. Full DRL implementation below.

plm-cfg-08-bundle-validation.drl

```

package com.sophix.plm.rules

import com.sophix.plm.facts.BundleFact
import com.sophix.plm.facts.BundleCompositionFact
import com.sophix.plm.facts.ExistingCodesFact
import com.sophix.plm.facts.PackageAvailabilityFact
import com.sophix.plm.facts.ReferencingSubscriptionsFact
import com.sophix.plm.facts.ValidationError
import com.sophix.plm.facts.ValidationWarning

// R-PLM-CFG-08-1 – Code unique across plm catalog (Services, Packages, Bundles share namespace)

rule "R-PLM-CFG-08-1-code-uniqueness"
when
    $b: BundleFact($code: code, $id: id)
    $existing: ExistingCodesFact(codes contains $code, ownerId != $id)
then
    insert(new ValidationError("R-PLM-CFG-08-1",
                                "code",
                                "Code already exists in catalog: " + $code));
end

// R-PLM-CFG-08-2 – Bundle must have at least 2 component Packages

rule "R-PLM-CFG-08-2-minimum-components"
when
    $b: BundleFact($id: id)
    $c: BundleCompositionFact(bundleId == $id, packages.size() < 2)

```

```

then
    insert(new ValidationError("R-PLM-CFG-08-2",
                              "composition",
                              "Bundle must contain at least 2 Packages; found " + $c.getPackages().size()));
end

// R-PLM-CFG-08-3 – On activate, all components must be Active

rule "R-PLM-CFG-08-3-components-must-be-active"
when
    $b: BundleFact(action == "ACTIVATE", $id: id)
    $pa: PackageAvailabilityFact(bundleId == $id, inactivePackages.size() > 0)
then
    insert(new ValidationError("R-PLM-CFG-08-3",
                              "composition",
                              "Cannot activate Bundle: component Packages not Active: " + $pa.getInactivePackages()));
end

// R-PLM-CFG-08-4 – Warning (not block) if bundle price > sum of components

rule "R-PLM-CFG-08-4-price-above-sum-warning"
when
    $b: BundleFact($id: id, $price: price)
    $pa: PackageAvailabilityFact(bundleId == $id, sumOfComponentPrices < $price)
then
    insert(new ValidationWarning("R-PLM-CFG-08-4",
                                "price",
                                "Bundle price (" + $price + ") exceeds sum of component prices (" + $pa.getSumOfComponentPrices() + ")"));
end

// R-PLM-CFG-08-5a – Bundle franchises must be subset of components' franchises

rule "R-PLM-CFG-08-5a-franchise-availability"
when
    $b: BundleFact($id: id)
    $pa: PackageAvailabilityFact(bundleId == $id, $franchisesNotCoveredByAllComponents: franchisesNotCoveredByAllComponents)
    eval(!$franchisesNotCoveredByAllComponents.isEmpty())
then
    insert(new ValidationError("R-PLM-CFG-08-5",
                              "targetFranchises",
                              "Bundle franchises must be subset of all components' franchises. Missing in some components: " + $pa.getFranchisesNotCoveredByAllComponents()));
end

// R-PLM-CFG-08-5b – Bundle tech regions must be subset of components' tech regions

rule "R-PLM-CFG-08-5b-tech-region-availability"
when
    $b: BundleFact($id: id)
    $pa: PackageAvailabilityFact(bundleId == $id, $techRegionsNotCoveredByAllComponents: techRegionsNotCoveredByAllComponents)
    eval(!$techRegionsNotCoveredByAllComponents.isEmpty())
then
    insert(new ValidationError("R-PLM-CFG-08-5",
                              "targetTechRegions",
                              "Bundle tech regions must be subset of all components' tech regions. Missing in some components: " + $pa.getTechRegionsNotCoveredByAllComponents()));
end

// R-PLM-CFG-08-6 – Soft-delete Bundle blocked if active subscriptions reference it

rule "R-PLM-CFG-08-6-delete-blocked-by-subscription"
when
    $b: BundleFact($id: id, action == "DELETE")
    $r: ReferencingSubscriptionsFact(bundleId == $id, activeSubscriptions.size() > 0)
then
    insert(new ValidationError("R-PLM-CFG-08-6",
                              "id",
                              "Bundle is referenced by active subscriptions: " + $r.getActiveSubscriptions().size() + " subscriptions"));
end

// R-PLM-CFG-08-7 – Package referenced by an active Bundle cannot be deleted
// (This rule fires when SIP-01 / PLM-CFG-01 attempts a Package delete and consults
// this rule package; see SIP-01 for the caller-side coordination.)

rule "R-PLM-CFG-08-7-package-delete-blocked-by-bundle"
when
    $p: BundleCompositionFact(action == "DELETE_PACKAGE", $packageId: packageId, $referencingActiveBundles: referencingActiveBundles)
    eval(!$referencingActiveBundles.isEmpty())
then
    insert(new ValidationError("R-PLM-CFG-08-7",
                              "packageId",
                              "Package " + $packageId + " is referenced by active Bundles: " + $referencingActiveBundles.size() + " bundles"));
end

```

```
// R-PLM-CFG-08-8 – Bundle billingFrequencyDays must be a positive integer (independent of components)

rule "R-PLM-CFG-08-8-billing-frequency-positive"
when
    $b: BundleFact($freq: billingFrequencyDays, action != "DELETE")
    eval($freq == null || $freq <= 0)
then
    insert(new ValidationError("R-PLM-CFG-08-8",
                              "billingFrequencyDays",
                              "billingFrequencyDays must be a positive integer; got " + $freq));
end
```

Fact classes (POJOs in com.sophix.plm.facts)

A Fact class is a plain Java object that Drools evaluates DRL rules against. The KJAR project (modules/plm-rules/) contains these classes; the rules engine deserializes JSON payloads (sent by the calling module over HTTPS) into these objects, fires rules, returns results as JSON. Facts use the standard `@Data` + `@NoArgsConstructor` + `@AllArgsConstructor` Lombok annotations (used throughout the codebase to remove getter/setter/constructor boilerplate).

BundleFact.java

```
package com.sophix.plm.facts;

import lombok.AllArgsConstructor;
import lombok.Data;
import lombok.NoArgsConstructor;

import java.math.BigDecimal;
import java.util.List;

@Data
@NoArgsConstructor
@AllArgsConstructor
public class BundleFact {
    private String id;
    private String code;
    private String name;
    private String status;
    private BigDecimal price;
    private String currency;
    private Integer billingFrequencyDays;
    private String defaultWalletRef;
    private String defaultTaxGroupRef;
    private List<String> targetFranchises;
    private List<String> targetTechRegions;
    private String action; // "CREATE" | "UPDATE" | "ACTIVATE" | "RETIRE" | "DELETE"
}
```

BundleCompositionFact.java

```
package com.sophix.plm.facts;

import lombok.AllArgsConstructor;
import lombok.Data;
import lombok.NoArgsConstructor;

import java.util.List;

@Data
@NoArgsConstructor
@AllArgsConstructor
public class BundleCompositionFact {
    private String bundleId;
    private List<String> packages; // packageIds in this bundle
    private String packageId; // for R-7 (Package-delete path)
    private List<String> referencingActiveBundles; // for R-7
    private String action; // "VALIDATE" | "DELETE_PACKAGE"
}
```

PackageAvailabilityFact.java

```
package com.sophix.plm.facts;

import lombok.AllArgsConstructor;
import lombok.Data;
import lombok.NoArgsConstructor;

import java.math.BigDecimal;
import java.util.List;

@Data
```

```

@NoArgsConstructor
@AllArgsConstructor
public class PackageAvailabilityFact {
    private String bundleId;
    private List<String> inactivePackages; // for R-3
    private BigDecimal sumOfComponentPrices; // for R-4
    private List<String> franchisesNotCoveredByAllComponents; // for R-5a
    private List<String> techRegionsNotCoveredByAllComponents; // for R-5b
}

```

ReferencingSubscriptionsFact.java

```

package com.sophix.plm.facts;

import lombok.AllArgsConstructor;
import lombok.Data;
import lombok.NoArgsConstructor;

import java.util.List;

@Data
@NoArgsConstructor
@AllArgsConstructor
public class ReferencingSubscriptionsFact {
    private String bundleId;
    private List<String> activeSubscriptions; // subscription IDs (string refs, no FK)
}

```

ValidationWarning.java (alongside ValidationError introduced in PLM-CFG-01)

```

package com.sophix.plm.facts;

import lombok.AllArgsConstructor;
import lombok.Data;
import lombok.NoArgsConstructor;

@Data
@NoArgsConstructor
@AllArgsConstructor
public class ValidationWarning {
    private String rule; // e.g., "R-PLM-CFG-08-4"
    private String field; // e.g., "price"
    private String message; // human-readable description
}

```

The caller's responsibility:

- For **create/update**: insert BundleFact + BundleCompositionFact (with the bundle's package list) + ExistingCodesFact (from catalog) + PackageAvailabilityFact (lookup of each component's status, price, and franchises); fire rules; collect ValidationError and ValidationWarning.
- For **activate**: same as create/update plus action="ACTIVATE" on the BundleFact to enable R-3.
- For **delete**: insert BundleFact (action="DELETE") + ReferencingSubscriptionsFact (from subscription-engine API or local view); fire rules; if ValidationError, reject with BUSINESS_RULE_ERROR carrying the rule ID.
- For **Package-delete** (coordinated with PLM-CFG-01 / SIP-01): the Package-delete path inserts BundleCompositionFact with action="DELETE_PACKAGE", packageId=... , and referencingActiveBundles=... to enable R-7.

ValidationWarning results are returned to the caller as advisory messages — the caller (UI) presents them as confirmation prompts rather than blocks.

Events (Kafka)

Event	Topic	When	Consumers
BundleCreated	sophix.plm.bundle.created	After successful insert	subscription-engine cache, fulfillment cache, billing cache, search index
BundleUpdated	sophix.plm.bundle.updated	After successful update	subscription-engine cache, fulfillment cache, billing cache, search index
BundleRetired	sophix.plm.bundle.retired	After soft-delete	subscription-engine cache, fulfillment cache, billing cache, search index

Key format: plm:bundle:{bundleId} · eventId format: plm:bundle:{bundleId}:evt_{ulid}.

BundleCreated envelope:

```
{
  "eventId": "plm:bundle:bnd_01HX2Y3Z4:evt_01JZ5K6...",
  "eventType": "BundleCreated",
  "aggregateId": "bnd_01HX2Y3Z4",
  "timestamp": "2026-05-15T10:23:00Z",
  "actor": "alex.oyuga",
  "correlationId": "corr_01HX...",
  "payload": {
    "bundleId": "bnd_01HX2Y3Z4",
    "code": "TRIPLE_PLAY_PRO",
    "name": "Triple Play Pro",
    "status": "DRAFT",
    "price": 5500,
    "currency": "KES",
    "billingFrequencyDays": 30,
    "defaultWalletRef": "TVD",
    "defaultTaxGroupRef": "Bundle-StandardVAT",
    "components": ["pkg_inet_fast_gpon", "pkg_tv_premium", "pkg_voice_fixed"],
    "targetFranchises": ["Nairobi", "Mombasa"],
    "targetTechRegions": ["GPON"]
  }
}
```

BundleUpdated envelope:

```
{
  "eventId": "plm:bundle:bnd_01HX2Y3Z4:evt_01JZ5L8...",
  "eventType": "BundleUpdated",
  "aggregateId": "bnd_01HX2Y3Z4",
  "timestamp": "2026-05-15T14:02:00Z",
  "actor": "alex.oyuga",
  "correlationId": "corr_01HX...",
  "payload": {
    "bundleId": "bnd_01HX2Y3Z4",
    "changedFields": {
      "price": { "from": 5500, "to": 5200 },
      "components": { "from": ["pkg_a", "pkg_b", "pkg_c"], "to": ["pkg_a", "pkg_b", "pkg_d"] }
    }
  }
}
```

BundleRetired envelope follows the PLM-CFG-01 retired pattern: just `bundleId` + `retiredAt`.

Cache strategy: `plm` does not cache its own data. Downstream caches use key prefix `sophix:plm:bundle:` with TTL 24h. See `FOUNDATION_CACHE.md`.

Exceptions

Standard exception taxonomy (reused across all DDs):

Code	Type	Triggered when
INPUT_VALIDATION_ERROR	Input validation	Required field missing, malformed JSON, type mismatch
BUSINESS_RULE_ERROR	Business rule	Any R-PLM-CFG-08-* rule rejects
BUSINESS_RULE_WARNING	Business rule warning	R-PLM-CFG-08-4 warning (returned alongside success unless user acknowledged)
INTERNAL_EXCEPTION	Internal	Unexpected runtime error
BACKEND_ERROR	Backend	DB unavailable, Kafka publish failure, KIE Server unreachable

Error response shape (same as PLM-CFG-01):

```
{ "error": "BUSINESS_RULE_ERROR", "rule": "R-PLM-CFG-08-2", "message": "...", "details": {...} }
```

APIs (in `plm/bundle-catalog/`)

Auth (per `FOUNDATION_AUTH.md`):

- From UI: Bearer JWT; role from claims enforces per-endpoint permissions
- From other modules: X-Service-Token header
- Either auth path accepted on read endpoints

Method + Path	Purpose	Auth role
POST /api/bundles	Create (in DRAFT state)	Product Manager, Admin

Method + Path	Purpose	Auth role
GET /api/bundles/{id}	Read	any authenticated
GET /api/bundles?status=&franchise=&page=&size=	List/filter	any authenticated
PATCH /api/bundles/{id}	Update	Product Manager, Admin
POST /api/bundles/{id}/activate	Transition DRAFT → ACTIVE	Product Manager, Admin
POST /api/bundles/{id}/retire	Transition ACTIVE → RETIRED	Product Manager, Admin
DELETE /api/bundles/{id}	Soft-delete (only if DRAFT or RETIRED with no active subs)	Product Manager, Admin
POST /api/bundles/preview-price-warning	UI helper: returns R-4 warning if price > sum	any authenticated

Status codes (same as PLM-CFG-01)

200 read/update · 201 create · 204 delete · 400 input validation · 401 unauth · 403 forbidden · 404 not found · 409 business rule conflict · 500 internal · 502/503/504 backend.

POST /api/bundles

Request:

```
{
  "code": "TRIPLE_PLAY_PRO",
  "name": "Triple Play Pro",
  "description": "Internet + TV + Voice combined",
  "price": 5500,
  "currency": "KES",
  "billingFrequencyDays": 30,
  "defaultWalletRef": "TVD",
  "defaultTaxGroupRef": "Bundle-StandardVAT",
  "componentPackageIds": ["pkg_inet_fast_gpon", "pkg_tv_premium", "pkg_voice_fixed"],
  "targetFranchises": ["Nairobi", "Mombasa"],
  "targetTechRegions": ["GPON"]
}
```

201 response:

```
{
  "id": "bnd_01HX2Y3Z4",
  "code": "TRIPLE_PLAY_PRO",
  "name": "Triple Play Pro",
  "status": "DRAFT",
  "price": 5500,
  "currency": "KES",
  "billingFrequencyDays": 30,
  "defaultWalletRef": "TVD",
  "defaultTaxGroupRef": "Bundle-StandardVAT",
  "componentPackageIds": ["pkg_inet_fast_gpon", "pkg_tv_premium", "pkg_voice_fixed"],
  "targetFranchises": ["Nairobi", "Mombasa"],
  "targetTechRegions": ["GPON"],
  "warnings": [],
  "createdAt": "2026-05-15T10:23:00Z",
  "createdBy": "alex.oyuga"
}
```

201 with warning (R-4 — price exceeds sum):

```
{
  "id": "bnd_01HX2Y3Z4",
  ...,
  "warnings": [
    {
      "rule": "R-PLM-CFG-08-4",
      "field": "price",
      "message": "Bundle price (7500) exceeds sum of component prices (6500). Confirm intentional."
    }
  ]
}
```

409 (R-1 — duplicate code):

```
{
  "error": "BUSINESS_RULE_ERROR",
  "rule": "R-PLM-CFG-08-1",
  "message": "Code already exists in catalog: TRIPLE_PLAY_PRO",
  "details": { "field": "code" }
}
```

400 (R-2 — too few components):

```
{
  "error": "BUSINESS_RULE_ERROR",
  "rule": "R-PLM-CFG-08-2",
  "message": "Bundle must contain at least 2 Packages; found 1",
  "details": { "field": "composition" }
}
```

POST /api/bundles/{id}/activate

No body. Fires R-3, R-5.

200 response: full updated Bundle (status now ACTIVE).

409 (R-3):

```
{
  "error": "BUSINESS_RULE_ERROR",
  "rule": "R-PLM-CFG-08-3",
  "message": "Cannot activate Bundle: component Packages not Active",
  "details": { "inactivePackages": ["pkg_tv_premium"] }
}
```

DELETE /api/bundles/{id}

Soft delete. Fires R-6.

204 on success.

409 (R-6 — active subscriptions reference this Bundle):

```
{
  "error": "BUSINESS_RULE_ERROR",
  "rule": "R-PLM-CFG-08-6",
  "message": "Bundle is referenced by 47 active subscriptions",
  "details": { "activeSubscriptionCount": 47 }
}
```

POST /api/bundles/preview-price-warning

UI helper for live calculation panel.

Request:

```
{
  "price": 5500,
  "componentPackageIds": ["pkg_inet_fast_gpon", "pkg_tv_premium", "pkg_voice_fixed"]
}
```

200 response:

```
{
  "sumOfComponentPrices": 6500,
  "implicitDiscount": 1000,
  "implicitDiscountPct": 15.38,
  "warnings": []
}
```

B — Current TZ

Bundle entity exists in TZ codebase (Bundle Java class in one of the billing modules) but:

- No formal lifecycle states (no Draft / Active / Retired transitions)
- No per-franchise availability (TZ is single-deployment)
- No composition rules in code (the bundle's package list is set in DB but not validated)
- No price-independence enforcement (the bundle price field exists but isn't separated semantically from sum of packages)
- No KRA tax-invoice integration
- No Kafka events on bundle changes
- Catalog admin via JHipster CRUD pages (basic, not workflow-aware)

The TZ Bundle entity is essentially a database row with no behavior — all business logic happens in adjacent code (subscription creation, billing).

C — Gap & bridge

Aspect	TZ today	Target	Bridge
Bundle entity	Java class, basic CRUD	Same shape, add status field + lifecycle methods	Reuse class, add status column, add transition methods
Composition	Stored as joined table	Same	Reuse
Pricing	Single price field	Same field; warning rule R-4 added	Add R-4 to rules KJAR; no schema change
Status / lifecycle	None	DRAFT / ACTIVE / RETIRED	Add status column; add activate/retire endpoints
Per-franchise availability	None	target_franchises array	New column; populate with default (all franchises) for existing bundles during migration
Validation rules	Hand-coded in services	Drools KJAR with 7 rules	New plm-cfg-08-bundle-validation.drl
Events	None	3 Kafka topics	New publishers using existing outbox pattern
Admin UI	JHipster pages	BSS Backoffice Catalog section	New screens; deprecate JHipster pages
Coordination with Package delete	None	R-7 fires from Package delete path	Coordinated with SIP-01 (Package management DD)

Dependencies

- PLM-CFG-01 (Service Catalog) — required first, since Packages compose Services. **Already completed.**
- SIP-01 (Package management) — required first or in parallel, since Bundles compose Packages. R-7 coordination required.
- Foundations (DB, Kafka, Drools, Cache, Auth) — all required. **Already in place.**